Group Membership & Partition Assignment

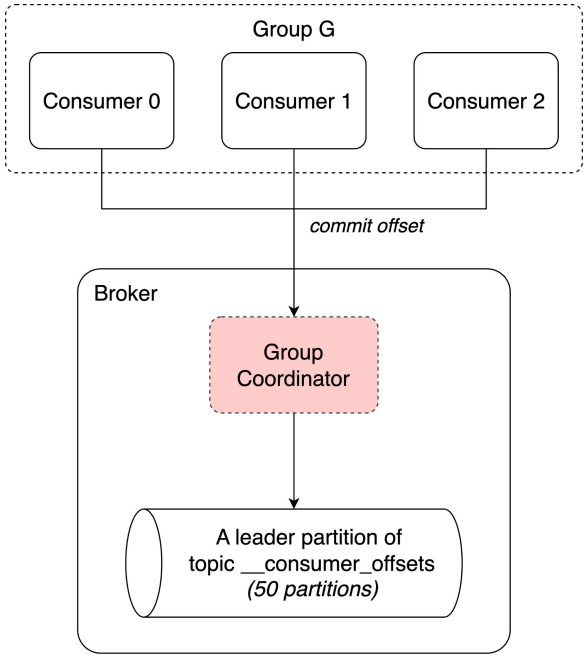
“Care about what other people think
and you will always be their prisoner.”
– Lao Tzu



1. Group Membership

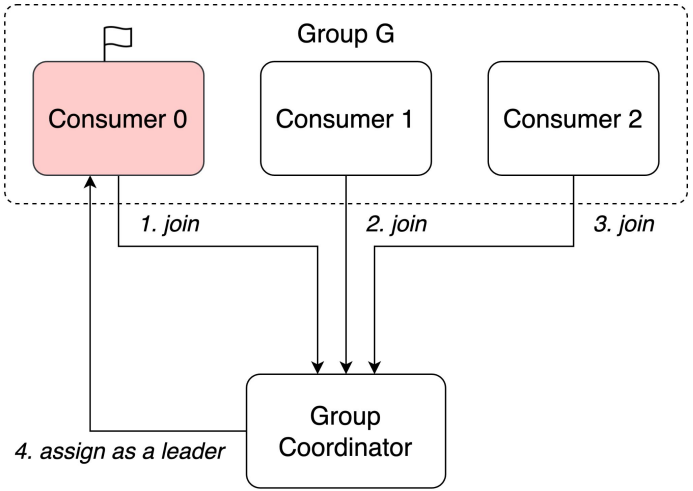
1.1. Group Coordinator

- A consumer group has a group coordinator.
- **The Group Coordinator is typically a process within a Kafka broker.**
- Functions:
  - Managing group membership
  - Partition assignment / Rebalancing
  - Tracking Offsets
  - Liveness, …
- A cluster has multiple group coordinators.
- A group coordinator serves multiple consumer consumer groups.



1.2. Group Leader

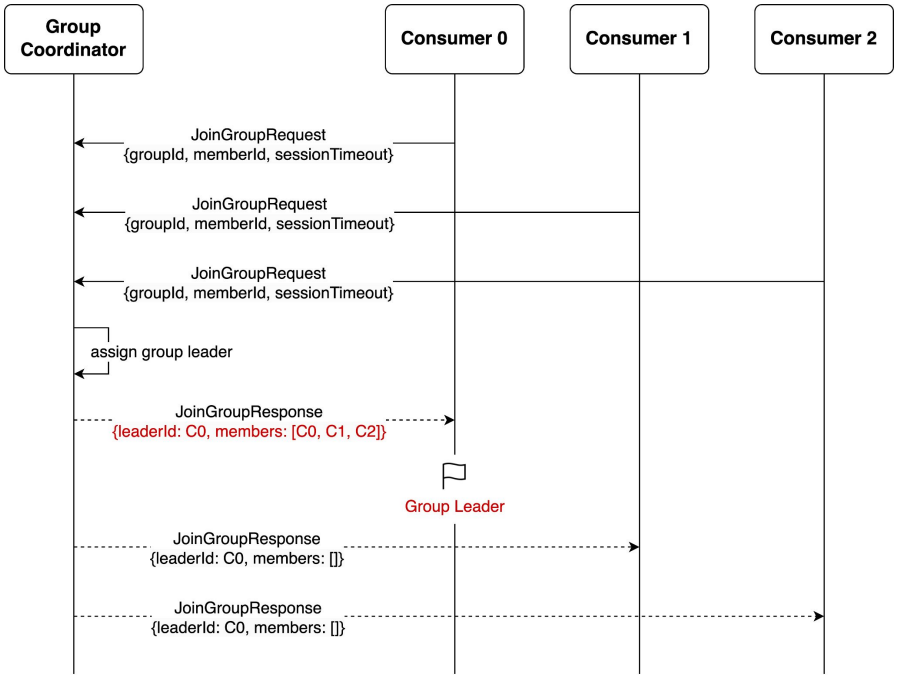
- Group Leader is a consumer and is assigned by the Group Coordinator.
- Functions:
  - **Determining partition assignment**
  - As a Consumer



2. Group Protocol

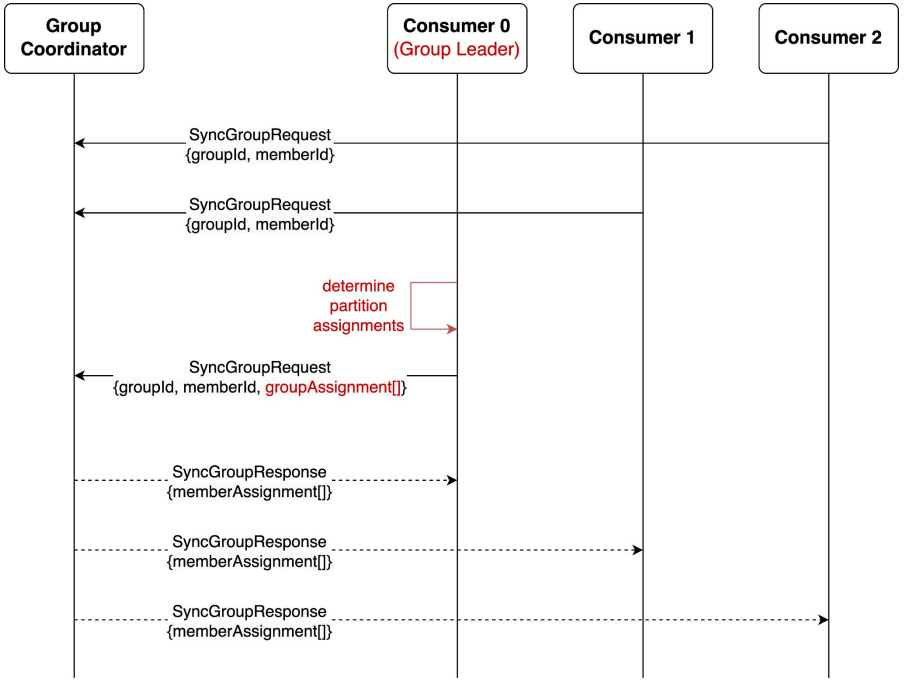
2.1. Membership Phrase

- Assume that 3 consumers join the same group simultaneously.
- The Group Coordinator assigns the group leader.
- **The Group Leader is usually the first consumer who joins the group.**



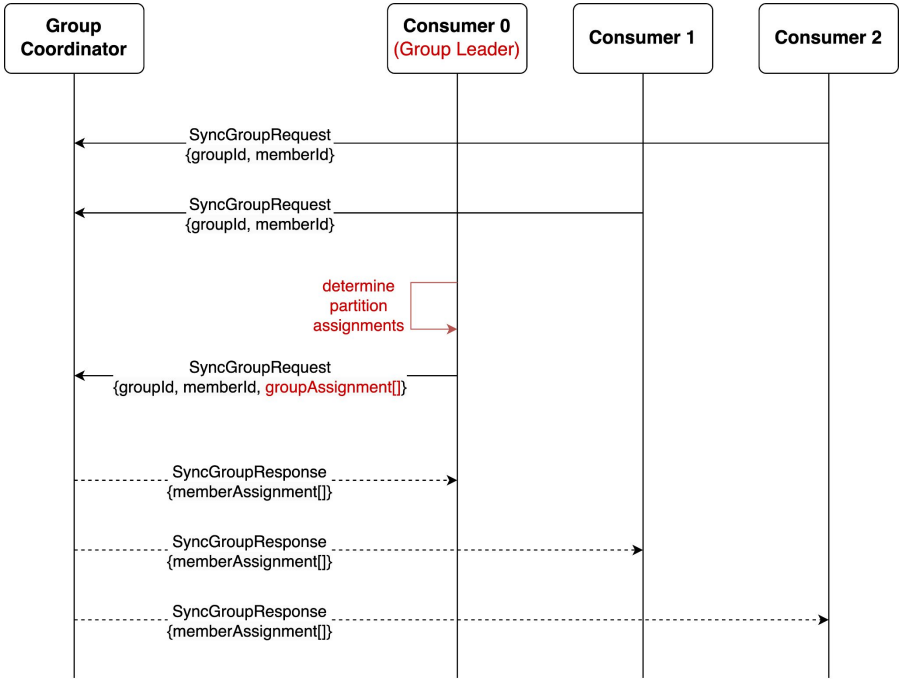
2.2. Synchronization Phrase

- **The Group Leader is responsible for determining partition assignments.**
- The Group Leader sends partition assignments to the Group Coordinator
- The Group Coordinator forwards partition assignments to the group members.



2.2. Synchronization Phrase

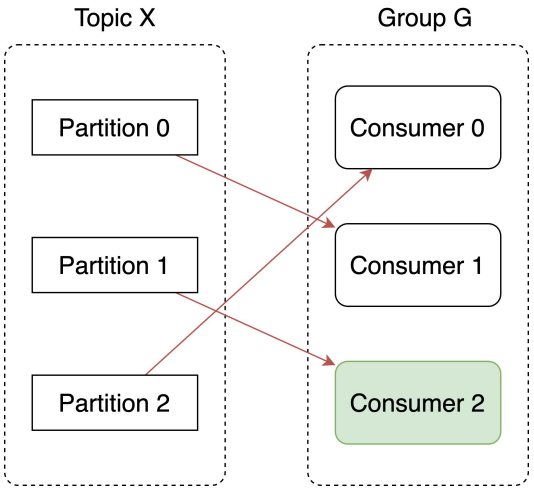
- Why Partition Assignment is determined on the consumer side?
- **Flexible to choose the partition assignment strategy.**



3. Eager Rebalance

3. Membership Phrase

- **All consumers in a group stop for a short period.**
- All partition assignments are revoked.
- The Group Lead determines a new partition assignment.
- All consumers join back to the group with the new assignment.
- After rebalance, **a consumer might be assigned to new partitions.**



4. Partition Assignment

4.1. Partition Assignment

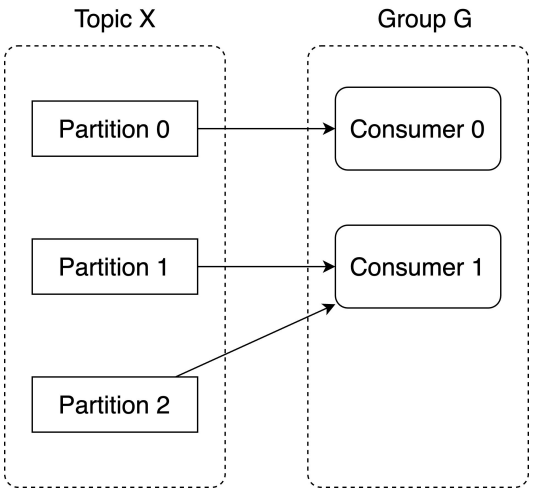
- What factors are considered in partition assignment?
- **Partition distribution**: the distribution of partitions across consumers, useful for balancing workloads.
- **Partition movement**: the number of reassigned partitions.
- **Eager**
- **Simplicity**

4.2. partition.assignment.strategy

	Pros	Cons
Range Assignor	● Simple and Predictable	● Eager ● High partition movement ● Uneven distribution for multiple topics
Round-Robin Assignor	● Even distribution for multiple topics	● Eager ● High partition movement in some cases
Sticky Assignor	● The low number of reassigned partitions	● Eager
Cooperative Sticky Assignor	● Not Eager ● The low number of reassigned partitions	● Complexity: maintaining the state when rebalancing

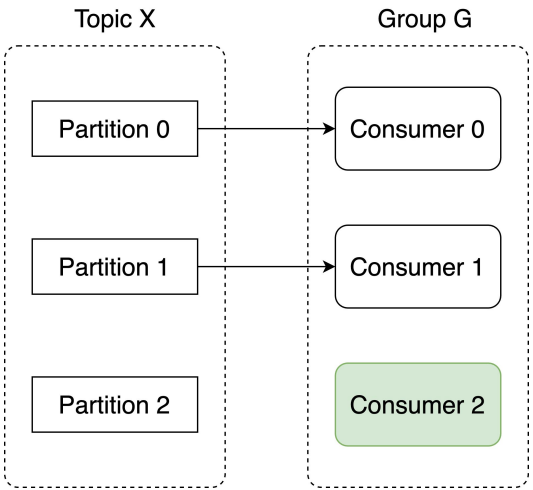
4.3. Cooperative Sticky Assignor

- Since Kafka 2.4.0



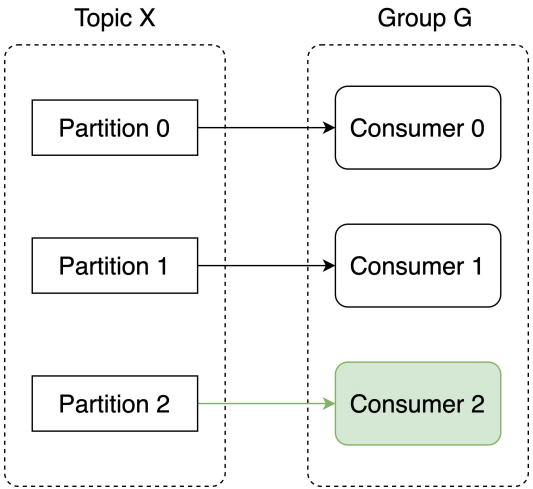
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- Pros:
  - **Reduce partition movement**
  - **Reduce the stop-the-world pause**
- Cons:
  - Complex



4.3. Cooperative Sticky Assignor

- Since Kafka 2.4.0
- Pros:
  - **Reduce partition movement**
  - **Reduce the stop-the-world pause**
- Cons:
  - Complex
- The default assignor:
  [RangeAssignor, CooperativeStickyAssignor]

